

Procedural Steps for Effective Delivery, Installation and Commissioning of Drilling Machine

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Received: February 2, 2014, Accepted: March 31, 2015, Published: March 31, 2015

ABSTRACT

Effective delivery and installation is important after proper selection of the machine tool in guaranteeing optimum performance of equipment while reducing fatalities. Besides, it ensures production starts off in the smoothest and most judicious way. In this study, drilling machine from the Central Workshop, College of Engineering, Afe Babalola University, Ado-Ekiti, Nigeria was used as a case study. Critical activities for effective installation and commissioning of the drilling machine were identified. The application and analysis of Critical Path Method (CPM) was employed in determining the values of Latest Start Time (LST), Earliest Start Time (EST), Processing Time (PT), Completion Time (CT) and float. The values were iteratively trained in a MATLAB R2010 environment with adequate Artificial Neural Network (ANN) using Marquardt Algorithm. The simulation results obtained were found to be very close compared to values obtained by manual calculations hence confirming conclusively that the model is a viable and predictive tool for effective machine delivery, commissioning and installation. The applications of the model is useful in machine delivery and installations in industries to ensure due dates are met while reducing tardiness.

Keyword: Artificial Neural Network, Completion time, Critical Path, Due Dates and Simulation Model

INTRODUCTION

The need to establish a effective procedure and practices for installations, commissioning and operation of machine tools so as to ensure optimum production with regards to quality, quantity and the environment have occupied the front burner due to machine tools installation and commissioning challenges ranging from tardiness during delivery, installation and commissioning stages to machine failure in service due to accidental falls, injuries or fatalities. Many industries have developed methods/procedure to minimise physical hazards while minimising machine downtime. The research work is in line with the yearning to develop best practices to meet equipment delivery, installation and commissioning due dates as well improve, quality, reliability and man-machine interaction. As manufacturers are under tremendous pressure to improve product quality in terms of dimension while maintaining high productivity; they need to address numerous problems in machine tools those affect the delivery, installation and commissioning activities. Solving or improving in all those problems areas is huge work but will provide the mean of evaluation of machine defects, potential safety and hazards and assist in the elimination of these hazards. This foster safe and healthy work environment via identification and control of risk associated with machine installation and commissioning. The work find its application in manufacturing industries as it could help gain a competitive

edge. Commissioning verifies the specified machine was installed; that it functions properly and it was successfully turned over to the user and reasonable ensures the next step verification for regulated industries would be successful [1].

It is a process by which an equipment, facility, or plant (which is installed, or is complete or near completion) is tested to verify if it functions according to its design objectives or specifications [1].

In practice, the commissioning process comprises the integrated application of a set of engineering techniques and procedures to check, inspect and test every operational component of the project, from individual functions, such as instruments and equipment, up to complex amalgamations such as modules, subsystems and systems. Next to efficiency, quality and delivery reliability have become key performance criteria [2] [3] cited by [4]. Predictive model of machine processes can be applied to help gain a competitive edge [5]

METHODOLOGY

Methods of optimizing due-date prediction and equipment procurement has been identified as network analysis, decision theory and empirical model [6]. Resilient back propagation is employed in training the network because its algorithm is one of the fastest for this purpose [7] [8] [9] [10] as cited by [11]

Table 1: Installation Activities for Drilling Machine

Activities	Precedence	Description	Duration (days)
A	-	Pre- installation activities	5
B	A	Arrival of the machine tools into the central workshop	1
C	B	Removal of the machines from the pallet boxes	3
D	B	Exercises involving check, and inspection of the machine tools	2
E	B	Use of transport equipment in moving machine tools to correct positions	1
F	C	Assembly of all subparts	2
G	E	Making connections	2
H	G	Testing all installation works	1
I	D F H	Proper check, inspection and testing activities	1
J	I	Completion of the relevant paper work	1

Table 2: Float Analysis for Drilling Machine Installation

Activities	LST	EST	Float	EFT (A)	LFT (B)	duration (m)	$E = \frac{a+4m+b}{6}$	$V = (\frac{b-a}{6})^2$
A	0	0	0	5	5	5	5	5
B	5	5	0	6	6	1	2.66	2.66
C	6	6	0	9	9	3	5	5
D	9	6	3	11	8	2	4.5	0.25
E	7	6	1	8	7	1	3.16	0.027
F	9	9	0	11	11	2	5	0
G	8	7	1	10	9	2	4.5	0.027
H	10	9	1	11	10	1	4.16	0.027
I	11	11	0	12	12	1	4.66	0
J	12	12	0	13	13	1	5	0

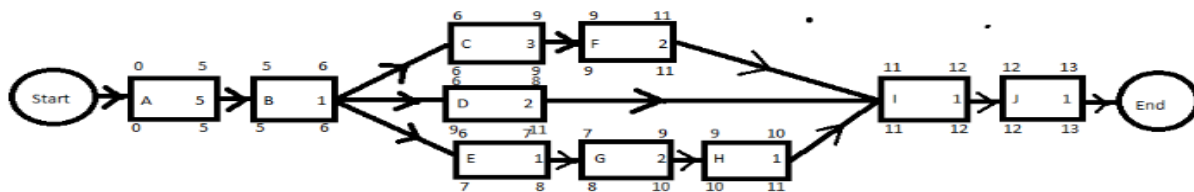


Fig. 1: Network Analysis for Drilling Machine Installation, Critical path A-B-C-F-I-J

Non- critical path D, E, G, H

Table 3: Drilling Machine Installation and Processing Time

Activities j	A	B	C	D	E	F	G	H	I	J
Processing time (days)	5	1	3	2	1	2	2	1	1	1

No of jobs: 10

Arrange the job as per SPT ordering [12]

Table 4: Drilling Machine Installation and Processing Time as Per SPT Ordering

Activities j	B	E	H	I	J	D	F	G	C	A
Processing time (days)	1	1	1	1	1	2	2	2	3	5

Therefore job sequence is as follows: B-E-H-I-J-D-F-G-C-A

Determination of Mean Flow Time

Table 5: Mean Flow Time for Drilling Machine Installation

Activities j	B	E	H	I	J	D	F	G	C	A
Processing time (days)	1	1	1	1	1	2	2	2	3	5
Completion time c_j	1	2	3	4	5	7	9	11	14	19

$$\bar{F} = \frac{1}{n} \sum_{i=1}^n f_j$$

$$\bar{F} = \frac{1}{10} \sum_{i=1}^{10} (1+2+3+4+5+7+9+11+14+19)$$

$$\bar{F} = \frac{1}{10}(7.5)$$

$$\bar{F} = 7.5 \text{ days}$$

Therefore, the optimal mean low time is 7.5 days

Table 6: Drilling Machine Installation and Due Time

Activities	A	B	C	D	E	F	G	H	I	J
Processing time t_j (days)	5	1	3	2	4	2	2	1	1	1
Due Date d_j (days)	5	3	5	5	3	5	5	4	5	5

The job as per EDD rule (i. e. increasing order of their due dates)

Table 7: Drilling Machine Installation and Completion Time

Jobs j	B	E	H	A	C	D	F	G	I	J
Due dates d_j (days)	3	3	4	5	5	5	5	5	5	5
Processing time t_j (days)	1	4	1	5	3	2	2	2	1	1
Completion time c_j (days)	1	5	6	11	14	16	18	20	21	22
Lateness l_j (days)	-2	-2	-2	6	9	11	13	15	16	17
Tardy/Non tardy (1/0)	0	1	1	1	1	1	1	1	1	1

The EDD sequence is B-E-H-A-C-D-F-G-I-J. From the last row in table 7, 0 means corresponding job is untardy and 1 means that the corresponding job is tardy. There are nine tardy jobs, the first is in the second column (k=2)

The maximum lateness value is 17, so this EDD sequence gives the minimum value for which is equal to 17. The application of Hodgson's algorithm gives a sequence which minimizes the total number of tardy jobs

Step 1: Place the sequence of jobs which is in EDD order in set E

$$E = E(B, E, H, A, C, D, F, G, I, J)$$

$$\text{set} L = (\text{Empty})$$

Step 2: Let

$$L = \{E\}$$

$$E = \{B, H, A, C, D, F, G, I, J\}$$

Table 8: Reduction of Tardy Jobs for Drilling Machine Installation

Jobs j	B	H	A	C	D	F	G	I	J
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Due dates d_j (days)	3	4	5	5	5	5	5	5	5
Processing time t_j (days)	1	1	5	3	2	2	1	1	1
Completion time c_j (days)	1	2	7	10	12	14	12	17	8
Lateness l_j (days)	-2	-2	2	5	7	6	9	12	13
Tardy/Non tardy (1/0)	0	0	1	1	1	1	1	1	1

It is clear that there are seven tardy jobs, the first tardy job is A which is in the third column (k=3) of the sequence in set E. Hence the number of tardy jobs should be minimized [13] [14]

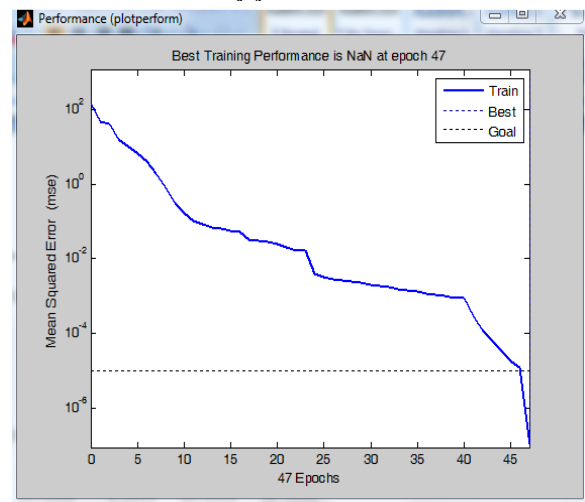


Fig. 2: Training Performance Graph

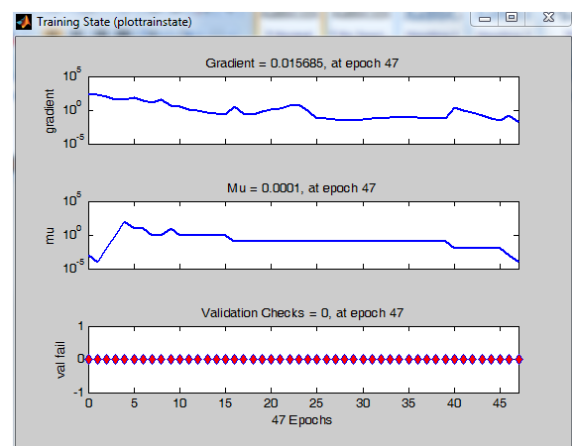


Fig. 3: Training State Graph

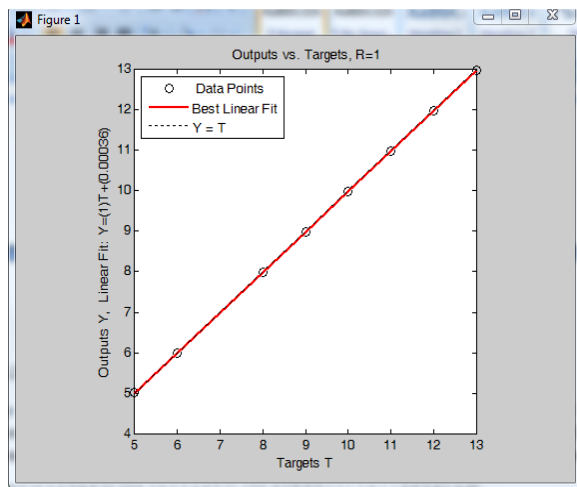


Fig. 4: Output Target Graph

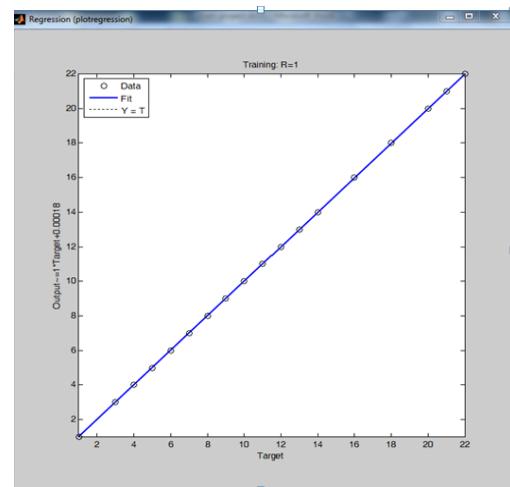


Fig. 5: Regression Graph

Table 9: Values of Completion Time and Due days Using Manual Calculation

Activities	LST (days)	EST (days)	Processing time (days)	Float	EFT (days)	LST (days)	Completion time (days)	Due Date (days)
B	5	5	1	0	6	6	1	5
E	7	6	4	1	8	7	5	3
H	10	9	1	1	11	10	6	5
A	0	0	5	0	5	5	11	5
C	6	6	3	0	9	9	14	3
D	9	6	2	3	11	8	16	5
F	9	9	2	0	11	11	18	5
G	8	7	2	1	10	9	20	4
I	11	11	1	0	12	12	21	5
J	12	12	1	0	13	13	22	5

Table 10: Values of Completion Time and Due days Using Artificial Neural Network

Activities	LST (days)	EST (days)	Processing time (days)	Float	EFT (days)	LST (days)	Completion time (days)	Due Date (days)
B	5	5	1	0	5.9998	6.0000	0.9995	4.9998
E	7	6	4	1	7.9998	7.0006	5.0002	3.0004
H	10	9	1	1	10.9997	1.0003	6.0001	5.0002
A	0	0	5	0	4.9995	5.0000	10.9999	5.0001
C	6	6	3	0	8.9994	8.9998	13.9995	3.0002
D	9	6	2	3	10.9997	8.0006	15.9997	5.0004
F	9	9	2	0	10.9998	10.9998	17.9997	5.0000
G	8	7	2	1	9.9996	9.9996	19.9993	4.0003
I	11	11	1	0	11.9996	11.9996	20.9998	5.0002
J	12	12	1	0	12.9997	12.9997	21.9999	5.0001

DISCUSSION OF RESULT

Table 2 shows that the developed network was very efficient with the line from the vertical axis cutting through its horizontal axis. The linearity of the output target graph (Fig. 4) and regression graph (Fig. 5) is an indication that predicted values were very close to the values calculated manually.

From table 9 and 10, the calculated values were very close to the predicted values for each of the selected machines. Thus, it can be concluded that the training of data set was successfully carried out and the model is a veritable tool for prediction of

completion time and due-date for effective delivery, installation and commissioning of machine tools.

From all the negligible values gotten from mean square error, it can be inferred that data were properly trained and the calculated values are not far from the predicted values.

CONCLUSION

- Activities involved in machine installation and commissioning were clearly identified;
- Critical paths for effective machine delivery, installation and commissioning were also identified and critical path method

was employed in predicting the due date for installation and commissioning of drilling machine;

c) The Marquardt algorithm was highly efficient for implementing the critical path method

d) The artificial neural network employed could closely predict values of due-dates having studied the interaction among imputed values

Recommendations

- 1) Emphasis should be made on the application of MATLAB software in Engineering courses, as it offers diverse solution to vast problems in the field.
- 2) The application of this research could be extended to other machine tools
- 3) The use of the predictive model by manufacturing industries could help gain a competitive edge

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Citation: A. O. Adeodu et al.,(2015) A. O. Adeodu et al. (2015). Procedural Steps for Effective Delivery, Installation and Commissioning of Drilling Machine. J. of Advancement in Engineering and Technology. V2I4. DOI: 10.15297/JAET.V2I4.02

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